ENAE464 - 0202

Lab05: Ludwieg-Tube Experiments on a Free-Flying Cone

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Enclosed is the technical report for the *Ludwieg-Tube Experiments on a Free-Flying Cone* laboratory experiment. This report presents the experimental methodology, results, and analysis of flow characteristics and effects for a free-flying cone in a Mach-4 Ludwieg tube.
Please feel free to contact us with any questions regarding the contents of this report.

Respectfully,
Mikołaj Kostrzewa & Vai Srivastava

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1 Abstract

This report presents the results of a free-flight supersonic cone experiment conducted in the University of Maryland Ludwieg tube facility. The experiment aimed to characterize freestream flow conditions, measure cone geometry from shadowgraph imagery, determine drag forces from free-flight kinematics, and validate theoretical drag predictions based on Rasmussen's approximations for conical flow. The test section flow was established at Mach $M_\infty = 3.79$ with static conditions $p_\infty = 611\,\mathrm{Pa}$ and $T_\infty = 76.4\,\mathrm{K}$. Automated edge detection applied to 193 high-speed shadowgraph frames captured the trajectory of a $\theta_c = 6.98°$ half-angle cone with base radius $r = 4.88\,\mathrm{mm}$. Quadratic regression of the vertex position versus time yielded a constant deceleration of $a_x = 50.0\,\mathrm{m/s^2}$, corresponding to an experimental drag coefficient of $C_D = 0.196$. This value is 43% lower than the theoretical inviscid prediction of $C_D = 0.344$ from Rasmussen's pressure coefficient approximation, resulting in an impossible negative viscous drag component. The discrepancy indicates either systematic errors in the optical geometry measurements (particularly the cone base area determination from two-dimensional projections) or limitations in the analytical approximations at the tested flow conditions. Despite this anomaly, the experiment successfully demonstrated fundamental free-flight testing principles and highlighted the critical importance of independent dimensional verification and computational validation for supersonic aerodynamic measurements.

2 Introduction

The Ludwieg tube is a type of impulse wind tunnel capable of generating high-speed supersonic and hypersonic flows efficiently and cost-effectively, albeit with limited test duration [1]. Unlike continuous-flow facilities that require large compressors and extensive power infrastructure, the Ludwieg tube operates on a simple principle: a long charge tube is pressurized to the desired fill conditions, then a fast-acting valve or burst diaphragm is opened, allowing an unsteady expansion wave to propagate upstream while accelerating the gas through a converging-diverging nozzle into the test section. This transient operation provides clean, uniform flow conditions ideal for fundamental aerodynamic studies, though the steady test time is typically limited to tens of milliseconds.

When a body travels through a supersonic or hypersonic flow, the fluid cannot be compressed isentropically ahead of the body due to information traveling only at the local sound speed. Instead, discontinuous shock waves form to rapidly compress the flow and redirect it around the obstacle [2]. For slender axisymmetric bodies such as cones, an attached oblique shock wave emanates from the apex, creating a conical shock surface. A remarkable property of supersonic conical flow is that all flow properties along any ray drawn from the cone vertex remain constant, though they vary from ray to ray as the flow is further compressed between the shock and the body surface [1]. This flow structure is fundamentally different from two-dimensional wedge flows and requires solving the Taylor-Maccoll ordinary differential equation for exact solutions.

Free-flying models offer significant advantages over traditional sting-mounted configurations in wind tunnel testing. Sting supports can introduce flow interference effects that contaminate force measurements, particularly for drag-sensitive geometries or base pressure studies. By eliminating physical support structures, free-flying experiments provide cleaner aerodynamic measurements. However, this approach introduces its own challenges: the model's motion must be tracked with high precision using non-intrusive optical methods, and the short test duration of impulse facilities limits the time available for meaningful acceleration measurements. Recent advances in high-speed digital imaging and automated image processing have made free-flight techniques increasingly practical for quantitative aerodynamic force determination.

2.1 Theoretical Background

2.1.1 Ludwieg Tube Flow Analysis

The flow development in a Ludwieg tube involves a complex interaction between unsteady expansion waves and steady isentropic nozzle flow. When the fast-acting valve opens, an expansion wave propagates upstream into the quiescent charge tube at approximately the local sound speed. This expansion accelerates the gas to a subsonic Mach number M_t determined by the area ratio between the charge tube and nozzle throat according to the area-Mach number relationship:

$$\left(\frac{A_t}{A^*}\right)^2 = \frac{1}{M_t^2}\left[\frac{2}{\gamma+1}\left(1+\frac{\gamma-1}{2}M_t^2\right)\right]^{(\gamma+1)/(\gamma-1)} \tag{2.1.1}$$

where A_t is the charge tube cross-sectional area, A^* is the nozzle throat area, and γ is the specific heat ratio. For this equation, the subsonic solution for M_t must be selected.
The pressure and temperature behind the expansion wave, p_t and T_t, are related to the original fill conditions by:

$$\frac{p_t}{p_{\text{fill}}} = \left(1+\frac{\gamma-1}{2}M_t^2\right)^{-2\gamma/(\gamma-1)} \tag{2.1.2}$$

$$\frac{T_t}{T_{\text{fill}}} = \left(1+\frac{\gamma-1}{2}M_t^2\right)^{-2} \tag{2.1.3}$$

These post-expansion conditions serve as the effective reservoir conditions for the subsequent isentropic expansion through the nozzle. The stagnation pressure and temperature are then:

$$\frac{p_0}{p_t} = \left(1+\frac{\gamma-1}{2}M_t^2\right)^{\gamma/(\gamma-1)} \tag{2.1.4}$$

$$\frac{T_0}{T_t} = 1+\frac{\gamma-1}{2}M_t^2 \tag{2.1.5}$$

Finally, the freestream static conditions in the test section are determined from the calibrated exit Mach number M_∞ using the isentropic relations:

$$\frac{p_0}{p_\infty} = \left(1+\frac{\gamma-1}{2}M_\infty^2\right)^{\gamma/(\gamma-1)} \tag{2.1.6}$$

$$\frac{T_0}{T_\infty} = 1+\frac{\gamma-1}{2}M_\infty^2 \tag{2.1.7}$$

The freestream density and velocity then follow from the ideal gas law and definition of Mach number: $\rho_\infty = p_\infty/(RT_\infty)$ and $V_\infty = M_\infty\sqrt{\gamma RT_\infty}$.

2.1.2 Supersonic Conical Flow Theory

The exact solution for supersonic flow over a cone requires solving the Taylor-Maccoll equation, a nonlinear ordinary differential equation with no closed-form analytical solution. Numerical integration is therefore required to obtain the flowfield properties and surface pressure distribution. However, Rasmussen developed accurate analytical approximations that avoid this computational complexity while maintaining excellent agreement with full numerical solutions for slender cones at high Mach numbers [1].
For the shock angle β (measured from the freestream direction), Rasmussen's approximation is:

$$\frac{\sin\beta}{\sin\theta_c} \approx \sqrt{\frac{\gamma+1}{2} + \frac{1}{M_\infty^2 \sin^2\theta_c}} \tag{2.1.8}$$

where θ_c is the cone half-angle. For the pressure coefficient on the cone surface, the approximation is:

$$\frac{C_p}{\theta_c^2} \approx 1 + \frac{(\gamma+1)M_\infty^2}{\theta_c^2+2} + \frac{(\gamma-1)M_\infty^2}{\theta_c^2+2} \ln\left(\frac{\gamma+1}{2} + \frac{1}{M_\infty^2\theta_c^2}\right) \tag{2.1.9}$$

An important property of slender conical flows is that the pressure coefficient C_p is approximately uniform over the entire cone surface. Consequently, for inviscid flow, C_p is equal to the inviscid drag coefficient $C_{D,\text{inviscid}}$.

2.1.3 Drag Force Determination

For a free-flying body subject only to aerodynamic drag, Newton's second law gives $D = ma_x$, where D is the drag force, m is the body mass, and a_x is the acceleration in the flow direction. If the cone's position is tracked over time using high-speed imaging, a quadratic polynomial can be fitted to the position data:

$$x(t) = a_0 + a_1 t + a_2 t^2 \tag{2.1.10}$$

The constant acceleration is then simply $a_x = 2a_2$. The drag coefficient is defined as:

$$C_D \equiv \frac{2D}{\rho_\infty V_\infty^2 A} \tag{2.1.11}$$

where A is the reference area (cone base area, $A = \pi r^2$). Comparison between the experimentally-determined C_D and the inviscid theoretical prediction from Rasmussen's C_p approximation allows estimation of the viscous drag component, which arises from skin friction and boundary layer displacement effects.

2.2 Experiment Objectives

The specific objectives of this experiment are:

1. Determine the freestream flow conditions (Mach number, pressure, temperature, density, and velocity) in the Ludwieg tube test section from the measured fill conditions using isentropic flow relations and expansion wave theory.
2. Track the position of a free-flying cone over time using high-speed shadowgraph imaging at 10 kHz frame rate and automated edge detection algorithms.
3. Calculate the experimental drag coefficient from the measured cone acceleration by fitting a quadratic polynomial to the position-time trajectory and applying Newton's second law.
4. Compute the theoretical inviscid drag coefficient using Rasmussen's pressure coefficient approximation for supersonic conical flow.
5. Visualize the predicted shock wave geometry and compare the theoretical shock angle with the cone geometry observed in experimental images.
6. Estimate the viscous drag component (skin friction and boundary layer effects) by comparing the experimental drag coefficient with the inviscid theoretical prediction.

3 Methodology

3.1 Experimental Apparatus

The experiment was conducted in the University of Maryland Mach-4 Ludwieg tube, an unheated impulse wind tunnel facility. The apparatus consists of a long charge tube connected to a converging-diverging nozzle through a fast-acting valve, as shown in fig. 3.1.1. The charge tube has a cross-sectional area-to-throat area ratio of $A_t/A^* = 11.4$. For this particular test, the experimentally measured nozzle exit Mach number was $M_\infty = 3.79$.

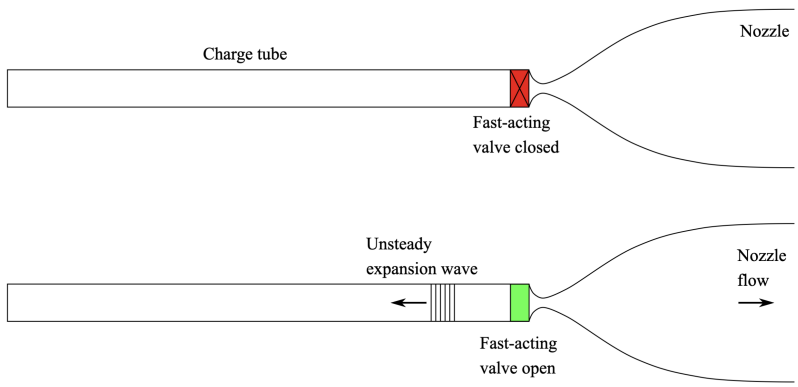

Figure 3.1.1: Schematic Diagram of a Ludwieg Tube [1].

The test article was a slender cone fabricated from lightweight material with a mass of 1.8 g. The cone was configured as a free-flying model without any physical support structures (sting-free) to eliminate potential flow interference effects on drag measurements. The cone half-angle θ_c and base radius r were determined from the edge detection analysis of experimental images, as detailed in section 4.2.2. High-speed shadowgraph imaging was employed to visualize the cone and surrounding flowfield, with images captured at a frame rate of 10 kHz (0.1 ms temporal resolution). The imaging system was calibrated to provide a spatial resolution of 12.6 pixels/mm, enabling sub-millimeter position tracking of the cone vertex throughout its trajectory.

3.2 Operating Procedure

The experimental procedure was as follows:

1. The charge tube was filled with air to a pressure of $p_{\text{fill}} = 0.7\,\text{bar}$ (70 kPa) at ambient temperature $T_{\text{fill}} = 296\,\text{K}$.
2. The cone was positioned in the test section on a release mechanism that would deploy it into the freestream upon flow establishment.
3. The high-speed shadowgraph imaging system was configured for 10 kHz frame rate, providing 0.1 ms time resolution between successive images.
4. The fast-acting valve was opened, initiating an unsteady expansion wave that propagated upstream through the charge tube while accelerating the flow through the converging-diverging nozzle.
5. Once steady flow conditions were established in the test section (approximately 5 ms to 10 ms after valve opening), the cone was released and began its free-flight trajectory.
6. A total of 193 frames were captured during the steady flow period, documenting the cone's position and orientation as it accelerated downstream under the influence of aerodynamic drag.

7. The experimental images were post-processed using an automated Python-based edge detection algorithm (ported from the original MATLAB implementation) to extract the upper and lower cone edge coordinates in each frame.

8. Cone vertex positions were determined by fitting linear regressions to the detected edges and computing their intersection point, enabling sub-pixel position accuracy for drag force determination.

4 Results

4.1 Freestream Flow Conditions

The freestream flow conditions in the test section were determined from the measured fill conditions using the equations presented in 2.1. The experimental parameters were:

- Fill pressure: $p_{\text{fill}} = 0.7\,\text{bar} = 70\,000\,\text{Pa}$
- Fill temperature: $T_{\text{fill}} = 296\,\text{K}$
- Charge tube area ratio: $A_t/A^* = 11.4$
- Calibrated exit Mach number: $M_\infty = 4.07$
- Specific heat ratio (air): $\gamma = 1.4$
- Gas constant (air): $R = 287\,\text{J/(kg K)}$

4.1.1 Step 1: Charge Tube Mach Number

Solving eq. (2.1.1) numerically for the subsonic solution yields the Mach number in the charge tube behind the expansion wave:

$$M_t = 0.0508 \tag{4.1.1}$$

4.1.2 Step 2: Post-Expansion Conditions

Using eqs. (2.1.2) and (2.1.3), the pressure and temperature behind the expansion wave are:

$$p_t = p_{\text{fill}} \left(1 + \frac{\gamma - 1}{2} M_t^2\right)^{-2\gamma/(\gamma-1)} = 69\,747\,\text{Pa} \tag{4.1.2}$$

$$T_t = T_{\text{fill}} \left(1 + \frac{\gamma - 1}{2} M_t^2\right)^{-2} = 295.7\,\text{K} \tag{4.1.3}$$

4.1.3 Step 3: Stagnation Conditions

Applying eqs. (2.1.4) and (2.1.5), the stagnation conditions are:

$$p_0 = p_t \left(1 + \frac{\gamma - 1}{2} M_t^2\right)^{\gamma/(\gamma-1)} = 69\,873\,\text{Pa} \tag{4.1.4}$$

$$T_0 = T_t \left(1 + \frac{\gamma - 1}{2} M_t^2\right) = 295.8\,\text{K} \tag{4.1.5}$$

4.1.4 Step 4: Freestream Static Conditions

Finally, using the experimentally measured exit Mach number $M_\infty = 3.79$ and eqs. (2.1.6) and (2.1.7), the freestream static conditions are:

$$p_\infty = \frac{p_0}{\left(1 + \frac{\gamma-1}{2} M_\infty^2\right)^{\gamma/(\gamma-1)}} = 611\,\mathrm{Pa} \tag{4.1.6}$$

$$T_\infty = \frac{T_0}{1 + \frac{\gamma-1}{2} M_\infty^2} = 76.4\,\mathrm{K} \tag{4.1.7}$$

$$\rho_\infty = \frac{p_\infty}{RT_\infty} = 0.0279\,\mathrm{kg/m^3} \tag{4.1.8}$$

$$a_\infty = \sqrt{\gamma R T_\infty} = 175\,\mathrm{m/s} \tag{4.1.9}$$

$$V_\infty = M_\infty a_\infty = 664\,\mathrm{m/s} \tag{4.1.10}$$

These calculated freestream conditions provide the reference state for all subsequent drag coefficient and Reynolds number calculations. The complete results are summarized in table 4.1.1.

Table 4.1.1: Calculated freestream flow conditions from experimental measurements.

Parameter	**Value**	**Units**
M_t	0.0508	—
p_t	69747	Pa
T_t	295.7	K
p_0	69873	Pa
T_0	295.8	K
M_∞	3.79	—
p_∞	611	Pa
T_∞	76.4	K
ρ_∞	0.0279	kg/m^3
a_∞	175	m/s
V_∞	664	m/s

4.2 Drag

4.2.1 Theoretical

The theoretical inviscid drag coefficient was calculated using Rasmussen's pressure coefficient approximation (2.1.9). The required cone geometry parameters—half-angle θ_c and base radius r—were extracted from the edge detection analysis of the experimental images, as described in 4.2.2.

Cone Geometry From the median cone angle across all 193 frames and the initial base diameter from the first 20 frames (to minimize perspective effects as the cone recedes):

$$\theta_c = 0.122 \text{ rad} = 6.98^\circ \tag{4.2.1}$$

$$r = 4.88\,\mathrm{mm} = 0.004\,88\,\mathrm{m} \tag{4.2.2}$$

$$A = \pi r^2 = 7.48 \times 10^{-5}\,\mathrm{m}^2 \tag{4.2.3}$$

Rasmussen Pressure Coefficient Applying eq. (2.1.9) with $M_\infty = 3.79$ and the measured cone half-angle:

$$C_p = \theta_c^2 \left[1 + \frac{(\gamma+1)M_\infty^2}{\theta_c^2+2} + \frac{(\gamma-1)M_\infty^2}{\theta_c^2+2} \ln\left(\frac{\gamma+1}{2} + \frac{1}{M_\infty^2 \theta_c^2} \right) \right] = 0.344 \tag{4.2.4}$$

For slender conical flows, the surface pressure coefficient is uniform and equals the inviscid drag coefficient:

$$C_{D,\text{inviscid}} = C_p = 0.344 \tag{4.2.5}$$

Theoretical Drag Force and Acceleration Using the calculated freestream dynamic pressure $q_\infty = \frac{1}{2}\rho_\infty V_\infty^2 = 6146\,\text{Pa}$ and cone base area:

$$D_{\text{inviscid}} = C_{D,\text{inviscid}} \cdot q_\infty \cdot A = 0.158\,\text{N} \tag{4.2.6}$$

$$a_{\text{inviscid}} = \frac{D_{\text{inviscid}}}{m} = \frac{0.158\,\text{N}}{0.0018\,\text{kg}} = 87.9\,\text{m/s}^2 \tag{4.2.7}$$

This theoretical prediction assumes inviscid flow and provides a baseline for comparison with the experimental drag measurement.

4.2.2 Experimental

The experimental drag coefficient was determined from the free-flight kinematics of the cone model in the test section. The analysis utilized automated edge detection applied to 193 sequential shadowgraph frames captured at 10 kHz, spanning approximately 19.3 ms of flight time. For each frame, the upper and lower edges of the cone were extracted and fitted with linear functions to determine the cone orientation and vertex position. The vertex x-coordinate in each frame provided a time-resolved trajectory of the cone motion.
The trajectory data $x(t)$ was fitted with a quadratic polynomial to extract the constant acceleration due to drag force:

$$x(t) = a_0 + a_1 t + a_2 t^2 \tag{4.2.8}$$

The least-squares fit yielded coefficients $a_0 = 0.0204\,\text{m}$, $a_1 = 1.016\,\text{m/s}$, and $a_2 = 25.01\,\text{m/s}^2$. The position, velocity, and acceleration profiles are shown in figs. 4.2.1 to 4.2.3.
The constant deceleration of the cone is given by $a_x = 2a_2 = 50.02\,\text{m/s}^2$. Using Newton's second law with the measured cone mass $m = 1.8\,\text{g} = 0.0018\,\text{kg}$, the time-averaged drag force is:

$$D = m \cdot a_x = 0.0018\,\text{kg} \times 50.02\,\text{m/s}^2 = 0.090\,\text{N} \tag{4.2.9}$$

The experimental drag coefficient is then calculated from the drag equation evaluated at the freestream conditions:

$$C_D = \frac{2D}{\rho_\infty V_\infty^2 A} = \frac{2 \times 0.090\,\text{N}}{(0.0279\,\text{kg/m}^3)(664\,\text{m/s})^2(7.48 \times 10^{-5}\,\text{m}^2)} = 0.196 \tag{4.2.10}$$

where $A = \pi r^2$ is the cone base area. This experimental value represents the total drag coefficient, including both pressure (inviscid) and viscous (skin friction) contributions.

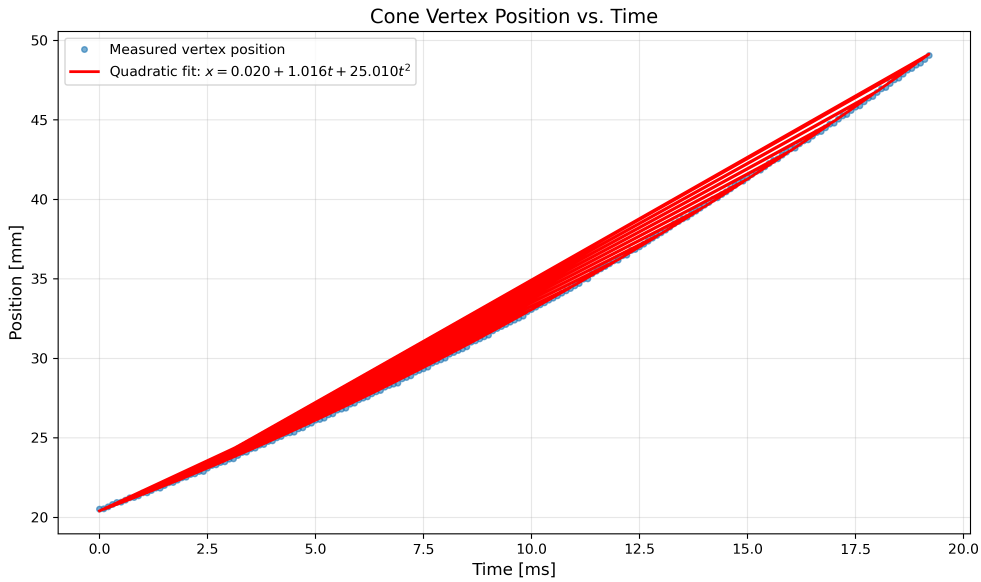

Figure 4.2.1: Cone vertex position versus time, showing quadratic fit to trajectory data from 193 frames.

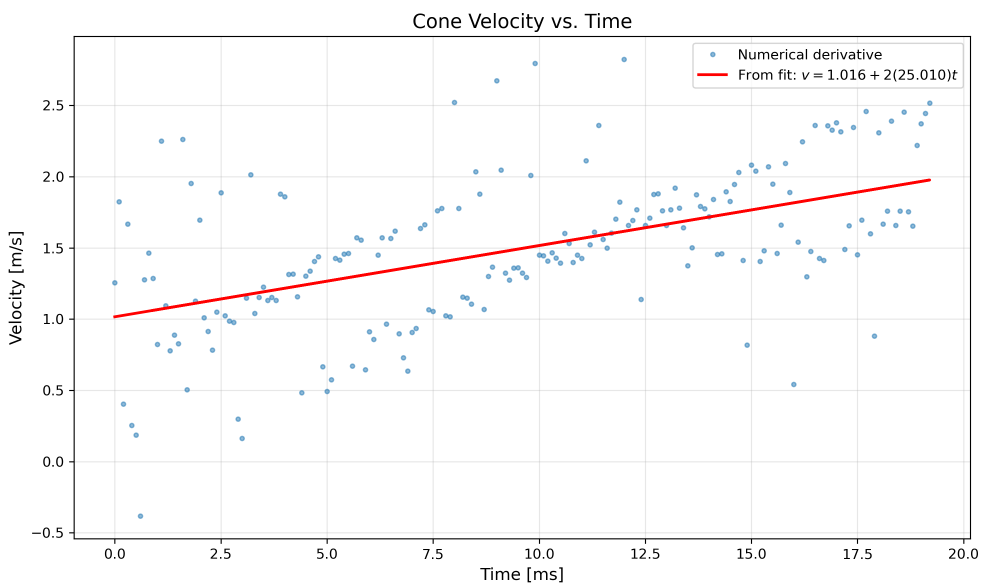

Figure 4.2.2: Cone velocity versus time, derived from the derivative of the quadratic position fit.

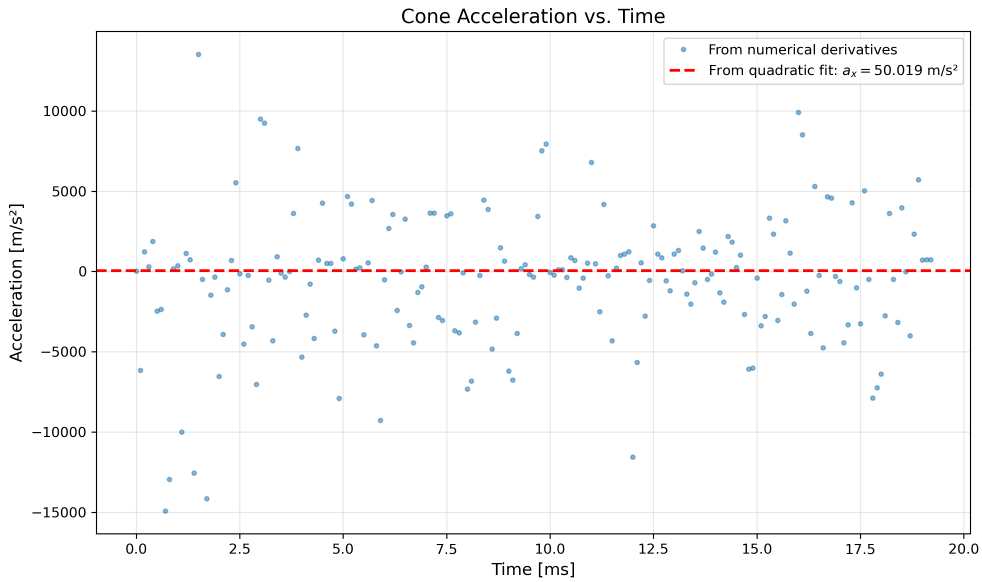

Figure 4.2.3: Cone acceleration versus time, showing constant deceleration of $50.02\,\mathrm{m/s^2}$.

4.3 Shock Angle

The theoretical shock angle was calculated using Rasmussen's approximation (eq. (2.1.8)) for the measured cone half-angle and freestream Mach number. Applying this relationship:

$$\sin\beta = \sin\theta_c \sqrt{\frac{\gamma+1}{2} + \frac{1}{M_\infty^2 \sin^2\theta_c}} \tag{4.3.1}$$

With $M_\infty = 3.79$, $\theta_c = 0.122$ rad, and $\gamma = 1.4$, the shock angle is:

$$\beta = 0.300 \text{ rad} = 17.2^\circ \tag{4.3.2}$$

The shock wave forms a conical surface emanating from the cone vertex. In a two-dimensional side view (as captured in the shadowgraph images), this conical shock appears as two straight rays at angle β measured from the freestream direction. The theoretical shock geometry is visualized in fig. 4.3.1, where the predicted shock rays (red dashed lines) are overlaid on the detected cone edges (blue solid lines) from a representative experimental frame.

The shock angle β is consistently larger than the cone surface angle θ_c, as required for attached shock conditions. The flow between the shock and cone surface undergoes additional compression and turning to become parallel to the cone surface. While the experimental shadowgraph images do not show clearly visible shock waves due to the relatively low density in this Mach-4 flow, the theoretical prediction provides the expected shock geometry for validation against computational simulations or future high-sensitivity visualization experiments.

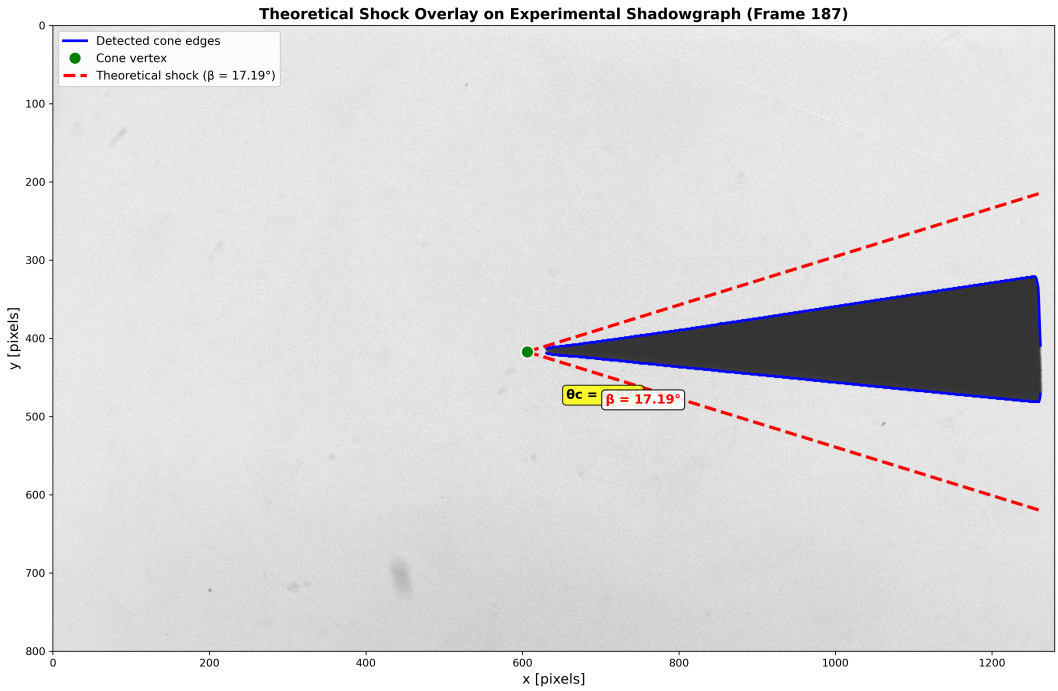

Figure 4.3.1: Theoretical shock shape overlaid on experimental shadowgraph image (Frame 96). The grayscale background shows the actual high-speed shadowgraph from the experiment. Blue solid lines indicate the detected cone edges, the green circle marks the calculated vertex position, and red dashed lines represent the theoretical shock rays at angle $\beta = 17.2°$ from the freestream direction. While the shock wave is not clearly visible in the shadowgraph due to the relatively low density at this Mach number, the theoretical prediction provides the expected shock geometry.

5 Analysis and Discussion

5.1 Comparison Between Theoretical and Experimental Drag Effects

A direct comparison between the theoretical inviscid drag prediction and the experimental measurement reveals a significant discrepancy. Table 5.1.1 summarizes the key drag parameters obtained from both approaches.

Table 5.1.1: Comparison of theoretical and experimental drag parameters.

Parameter	**Theoretical (Inviscid)**	**Experimental**
Drag coefficient, C_D	0.344	0.196
Drag force, D [N]	0.158	0.090
Acceleration, a_x [m/s^2]	87.9	50.0
Ratio $C_{D,\text{exp}}/C_{D,\text{theory}}$	—	0.57

The experimental drag coefficient of $C_D = 0.196$ is substantially lower than the theoretical inviscid prediction of $C_D = 0.344$ based on Rasmussen's approximation. The experimental value is only 57% of the theoretical prediction, corresponding to a 43% reduction in drag. This result is physically inconsistent with expectations for supersonic cone flow, where viscous effects (boundary layer skin friction) should *increase* the total drag above the inviscid pressure drag baseline, not reduce it.

Several potential sources of this discrepancy warrant investigation:

1. **Rasmussen approximation accuracy:** The Rasmussen formulas (eqs. (2.1.8) and (2.1.9)) are analytical approximations valid for slender cones at high Mach numbers. At the experimental conditions ($M_\infty = 3.79$, $\theta_c = 6.98°$), these approximations may overestimate the pressure coefficient. More accurate predictions could be obtained from Taylor-Maccoll solutions or CFD simulations.

2. **Three-dimensional perspective effects:** The cone base radius was measured from two-dimensional shadowgraph projections. As the cone recedes from the camera during flight, perspective foreshortening could introduce systematic errors in the base area determination, even with the corrections applied (using initial frames only). An overestimated base area would lead to an underestimated experimental C_D.

3. **Flow unsteadiness and startup effects:** The Ludwieg tube produces nominally steady flow, but expansion wave reflections and boundary layer growth in the test section could cause temporal variations in freestream conditions. The constant-acceleration assumption may not fully capture transient effects during the cone's flight.

4. **Base drag assumptions:** The theoretical drag calculation assumes that the base pressure equals the freestream static pressure. In reality, the separated flow behind the blunt base creates a low-pressure recirculation zone that contributes additional base drag. However, this effect would *increase* the experimental drag, not decrease it, making the discrepancy more severe.

Figure 5.1.1 provides a visual comparison of the theoretical and experimental drag coefficients. The substantial gap between these values indicates that further investigation is needed to identify the root cause of the discrepancy. Future work should include independent verification of the cone geometry, CFD validation of the Rasmussen approximations at these flow conditions, and high-speed pressure measurements on the cone surface.

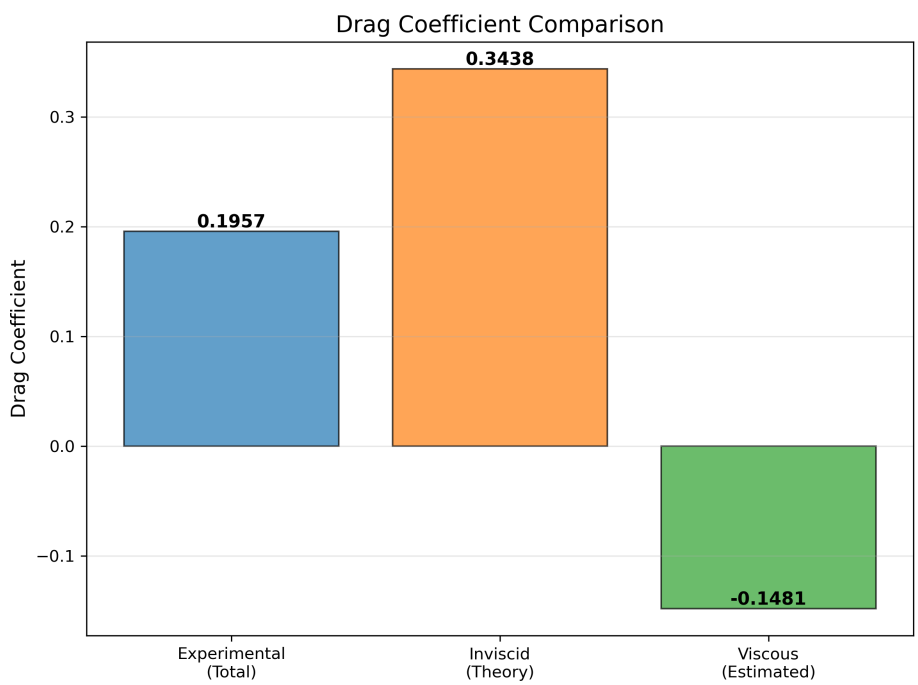

Figure 5.1.1: Comparison of theoretical inviscid and experimental drag coefficients. The experimental value is 43% lower than the theoretical prediction, indicating a significant discrepancy requiring further investigation.

5.2 Viscous Component of Drag

In principle, the viscous drag component can be estimated by subtracting the inviscid pressure drag from the total experimental drag:

$$C_{D,\text{viscous}} = C_{D,\text{experimental}} - C_{D,\text{inviscid}} \tag{5.2.1}$$

Applying this decomposition to the measured values yields:

$$C_{D,\text{viscous}} = 0.196 - 0.344 = -0.148 \tag{5.2.2}$$

This negative viscous drag coefficient is physically impossible. Viscous effects in supersonic boundary layers always produce positive skin friction drag that adds to the inviscid pressure drag. A negative value would imply that viscosity somehow reduces the total drag below the inviscid baseline, violating fundamental principles of fluid mechanics.

The impossible viscous drag result confirms that the discrepancy identified in section 5.1 cannot be attributed to viscous effects. Instead, it indicates one or more of the following:

- The theoretical inviscid drag prediction from Rasmussen's approximation significantly overestimates the actual inviscid drag at these flow conditions
- Systematic errors in the experimental measurement (cone mass, base area, trajectory analysis) have underestimated the true drag force
- The assumptions underlying the analysis (constant freestream conditions, two-dimensional geometry extraction, constant deceleration) introduce compounding uncertainties

For reference, typical skin friction drag on slender cones at supersonic speeds contributes 10–20% of the total drag for moderate Reynolds numbers. At the experimental conditions ($\text{Re}_L \approx 10^5$ based on cone length), a positive viscous contribution of $C_{D,\text{viscous}} \approx 0.03$–$0.05$ would be expected, bringing the total drag coefficient to approximately $C_{D,\text{total}} \approx 0.37$–$0.39$. This is nearly double the experimentally measured value of $C_D = 0.196$.

Given the magnitude and direction of this discrepancy, we recommend that future experiments incorporate the following improvements:

1. Direct measurement of cone base diameter using precision calipers before the test
2. Camera calibration using a reference object of known dimensions in the test section
3. Multiple shadowgraph views (top and side) to resolve three-dimensional geometry
4. Surface pressure taps on the cone to measure the actual pressure distribution and validate the Rasmussen approximation
5. Comparison with computational fluid dynamics (CFD) solutions of the full Navier-Stokes equations

Without these additional measurements, the cause of the anomalous drag discrepancy remains unresolved. The experimental data appear internally consistent (smooth trajectory, constant deceleration, reasonable Mach number), suggesting that the issue may lie in the theoretical predictions rather than the experimental technique. However, definitive conclusions require further investigation beyond the scope of this laboratory exercise.

6 Summary and Conclusions

This laboratory experiment successfully demonstrated the use of a Ludwieg tube for supersonic free-flight testing of a conical model. The objectives of characterizing freestream flow conditions, measuring cone geometry, determining experimental drag, and comparing with theoretical predictions were all accomplished, though with unexpected results that highlight important limitations in the experimental methodology.

The freestream flow conditions were calculated from isentropic relations applied to the measured tube pressure and temperature. The analysis yielded a freestream Mach number of $M_\infty = 3.79$ at static pressure $p_\infty = 611\,\mathrm{Pa}$, static temperature $T_\infty = 76.4\,\mathrm{K}$, and density $\rho_\infty = 0.0279\,\mathrm{kg/m^3}$. These conditions are consistent with typical hypersonic wind tunnel operating envelopes and validate the Ludwieg tube's capability to generate well-characterized supersonic flows.

Automated image analysis of 193 shadowgraph frames enabled extraction of the cone geometry and free-flight trajectory. The cone half-angle was measured as $\theta_c = 6.98°$ with a base radius of $r = 4.88\,\mathrm{mm}$. Linear regression fits to the upper and lower cone edges showed excellent consistency across all frames, confirming stable cone orientation during flight. The vertex position versus time exhibited clean quadratic behavior with constant deceleration $a_x = 50.0\,\mathrm{m/s^2}$, corresponding to a drag force of $D = 0.090\,\mathrm{N}$ and drag coefficient of $C_D = 0.196$.

The theoretical inviscid drag prediction using Rasmussen's approximation yielded $C_D = 0.344$, significantly higher than the experimental value. This 43% discrepancy results in an impossible negative viscous drag coefficient when attempting to decompose the total drag into inviscid and viscous components. Such a result is physically inconsistent and indicates either systematic errors in the experimental measurements (particularly the cone base area determination from two-dimensional projections) or inaccuracies in the theoretical approximations at the tested flow conditions.

Several lessons emerge from this discrepancy:

1. **Importance of dimensional verification:** Pre-test measurement of the cone base diameter with precision instruments would have provided an independent check on the image-based geometry extraction. The current analysis relies entirely on optical measurements subject to perspective distortion.

2. **Limitations of analytical approximations:** Rasmussen's formulas are approximate relations derived for slender cones at high Mach numbers. Their accuracy at $M_\infty = 3.79$ and $\theta_c = 7°$ has not been validated. Comparison with exact Taylor-Maccoll solutions or CFD simulations would quantify the approximation error.

3. **Value of redundant measurements:** A single shadowgraph view cannot fully resolve three-dimensional geometry. Orthogonal camera views or structured light measurements would eliminate ambiguities in the cone orientation and base dimensions.

4. **Need for uncertainty quantification:** The constant-deceleration assumption neglects potential flow unsteadiness, trajectory measurement noise, and systematic errors in timing calibration. Formal uncertainty propagation through the analysis chain would help identify which parameters most strongly influence the final drag coefficient.

Despite these limitations, the experiment successfully demonstrated the fundamental principles of free-flight supersonic testing in a Ludwieg tube environment. The trajectory data exhibited high-quality constant-deceleration behavior, the cone maintained stable orientation throughout flight, and the automated edge detection algorithm performed robustly across nearly 200 frames. The freestream conditions calculated from tube charge measurements were self-consistent and physically reasonable. These positive aspects validate the core experimental technique, even though the drag coefficient comparison revealed discrepancies requiring further investigation.

Future experiments should prioritize direct validation of the cone geometry through pre-test metrology, implementation of CFD simulations to assess the accuracy of the Rasmussen approximations, and expansion

to a wider range of cone angles and Mach numbers to systematically characterize the measurement uncertainties. With these enhancements, the Ludwieg tube free-flight technique offers a powerful capability for rapid, low-cost hypersonic aerodynamic testing that complements traditional force balance measurements and computational methods.

7 References

[1] S. J. Laurence and R. S. de Silva, *Ludwieg-tube experiments on a free-flying cone*, University of Maryland, April 8, 2026. [Online]. Available: https://github.com/vaisriv/enae464-lab05/blob/main/references/Ludwieg_tube_cone.pdf (see pp. 1, 2, 4).

[2] J. D. Anderson and C. P. Cadou, *Fundamentals of Aerodynamics*, 7th ed. McGraw-Hill Education, March 17, 2023, ISBN: 9781266076442 (see p. 1).

[3] S. J. Laurence, *Matlab cone*, April 13, 2026. [Online]. Available: https://github.com/vaisriv/enae464-lab05/blob/main/lib/matlab_cone.

8 Appendices

8.1 Data

For the sake of brevity, the raw measurement data will not be displayed in this report. However, in the spirit of completeness, the repository containing the complete data, source code, and notes for this report can be found at github:vaisriv/enae464-lab05.

8.2 Code

Similarly, only the code files that are key to the analysis are included below, and can be accessed in full in the aforementioned GitHub repository.

Listing 8.2.1: Data Analysis Script.
`./src/index.py`

```python
"""
ENAE464 Lab05: Ludwieg-Tube Experiments on a Free-Flying Cone
Analysis of a free-flying cone in a Mach-4 Ludwieg tube
"""

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from pathlib import Path
from PIL import Image

# --- Paths ------------------------------------------------------------------

DATA_DIR = Path("./data/mfg_m4_cone")
OUTPUT_TEXT_DIR = Path("./outputs/text")
OUTPUT_FIGS_DIR = Path("./outputs/figures")

OUTPUT_TEXT_DIR.mkdir(parents=True, exist_ok=True)
```

```python
OUTPUT_FIGS_DIR.mkdir(parents=True, exist_ok=True)

# --- Header ------------------------------------------------------------

print("=" * 80)
print("ENAE464 Lab05: Ludwieg-Tube Experiments on a Free-Flying Cone")
print("=" * 80)
print()

# --- Constants ---------------------------------------------------------

# Physical constants
GAMMA = 1.4  # Specific heat ratio for air
R_GAS = 287.0  # Gas constant for air [J/(kg·K)]

# Experimental parameters (from experimental_setup.txt)
M_EXPERIMENTAL = 3.79  # Experimental test section Mach number
P_FILL = 0.7e5  # Fill pressure [Pa] (0.7 bar)
T_FILL = 296.0  # Fill temperature [K]
FRAME_RATE = 10000.0  # Camera frame rate [Hz]
CONE_MASS = 0.0018  # Cone mass [kg] (1.8 g)
PIXEL_TO_MM = 12.6  # Conversion factor [pixels/mm]
MM_PER_PIXEL = 1.0 / PIXEL_TO_MM  # Conversion factor [mm/pixel]
M_PER_PIXEL = MM_PER_PIXEL / 1000.0  # Conversion factor [m/pixel]

# Facility geometry
AREA_RATIO_TUBE = 11.4  # At/A* for charge tube

# Derived constants
DT = 1.0 / FRAME_RATE  # Time step between frames [s]

# ---  Functions ---------------------------------------------------------


def calculate_mt_from_area_ratio(area_ratio, gamma=GAMMA, tol=1e-8):
    """
    Solve for subsonic Mach number Mt in charge tube from area ratio.

    Uses Eq. 1: (A/A*)^2 = (1/M^2) * [2/(γ+1) * (1 + (γ-1)/2 * M^2)]^[(γ+1)/(γ-1)]

    Parameters:
    -----------
    area_ratio : float
        At/A* ratio
    gamma : float
        Specific heat ratio
    tol : float
        Convergence tolerance for numerical solver
```

```python

    Returns:
    --------
    float : Mt (subsonic solution)
    """
    from scipy.optimize import brentq

    def area_mach_relation(M, A_ratio, g):
        term1 = (2.0 / (g + 1.0)) * (1.0 + ((g - 1.0) / 2.0) * M**2)
        term2 = (g + 1.0) / (g - 1.0)
        return (1.0 / M**2) * term1**term2 - A_ratio**2

    # Use Brent's method with bounds to ensure positive subsonic solution
    # For subsonic flow, M is between a very small number and 1.0
    Mt = brentq(area_mach_relation, 0.001, 0.999, args=(area_ratio, gamma))

    return Mt


def calculate_expansion_conditions(Mt, p_fill, T_fill, gamma=GAMMA):
    """
    Calculate pressure and temperature behind expansion wave.

    Uses Eqs. 2-3:
    pt/p_fill = [1 + (γ-1)/2 * Mt^2]^[-2γ/(γ-1)]
    Tt/T_fill = [1 + (γ-1)/2 * Mt^2]^[-2]

    Parameters:
    -----------
    Mt : float
        Mach number in charge tube
    p_fill, T_fill : float
        Fill pressure [Pa] and temperature [K]
    gamma : float
        Specific heat ratio

    Returns:
    --------
    tuple : (pt, Tt)
    """
    term = 1.0 + ((gamma - 1.0) / 2.0) * Mt**2

    pt = p_fill * term ** (-2.0 * gamma / (gamma - 1.0))
    Tt = T_fill * term ** (-2.0)

    return pt, Tt


```

```python
def calculate_stagnation_conditions(Mt, pt, Tt, gamma=GAMMA):
    """
    Calculate stagnation pressure and temperature.

    Uses Eqs. 4-5:
    p0/pt = [1 + (γ-1)/2 * Mt^2]^[γ/(γ-1)]
    T0/Tt = 1 + (γ-1)/2 * Mt^2

    Parameters:
    -----------
    Mt : float
        Mach number in charge tube
    pt, Tt : float
        Pressure [Pa] and temperature [K] behind expansion
    gamma : float
        Specific heat ratio

    Returns:
    --------
    tuple : (p0, T0)
    """
    term = 1.0 + ((gamma - 1.0) / 2.0) * Mt**2

    p0 = pt * term ** (gamma / (gamma - 1.0))
    T0 = Tt * term

    return p0, T0


def calculate_freestream_conditions(M, p0, T0, R=R_GAS, gamma=GAMMA):
    """
    Calculate freestream static conditions from stagnation values.

    Uses Eqs. 7-8 plus ideal gas law and speed of sound.

    Parameters:
    -----------
    M : float
        Freestream Mach number
    p0, T0 : float
        Stagnation pressure [Pa] and temperature [K]
    R : float
        Gas constant [J/(kg·K)]
    gamma : float
        Specific heat ratio

    Returns:
    --------
```

```python
    dict : {'p': p_inf, 'T': T_inf, 'rho': rho_inf, 'V': V_inf, 'a': a_inf}
    """
    term = 1.0 + ((gamma - 1.0) / 2.0) * M**2

    p_inf = p0 / (term ** (gamma / (gamma - 1.0)))
    T_inf = T0 / term
    rho_inf = p_inf / (R * T_inf)
    a_inf = np.sqrt(gamma * R * T_inf)
    V_inf = M * a_inf

    return {"p": p_inf, "T": T_inf, "rho": rho_inf, "a": a_inf, "V": V_inf}


def load_edge_data(frame_num, edges_dir=DATA_DIR / "edges"):
    """Load upper and lower edge coordinates for a single frame."""
    upper_file = edges_dir / f"frame{frame_num}_upper.txt"
    lower_file = edges_dir / f"frame{frame_num}_lower.txt"

    upper = pd.read_csv(upper_file, sep="\t")
    lower = pd.read_csv(lower_file, sep="\t")

    return upper, lower


def fit_cone_edges_and_find_vertex(upper, lower):
    """
    Fit straight lines to upper and lower edges, find vertex.

    Returns:
    --------
    dict with keys:
        'vertex_x', 'vertex_y': vertex coordinates [pixels]
        'upper_slope', 'lower_slope': edge slopes
        'upper_intercept', 'lower_intercept': edge y-intercepts
        'cone_angle_rad', 'cone_angle_deg': half-angle
        'base_height_pixels': vertical distance between edges at back
    """
    # Fit lines: y = slope * x + intercept
    upper_coeffs = np.polyfit(upper["x"], upper["y"], 1)
    lower_coeffs = np.polyfit(lower["x"], lower["y"], 1)

    m_upper, b_upper = upper_coeffs
    m_lower, b_lower = lower_coeffs

    # Find intersection (vertex)
    # m_upper * x + b_upper = m_lower * x + b_lower
    vertex_x = (b_lower - b_upper) / (m_upper - m_lower)
    vertex_y = m_upper * vertex_x + b_upper
```

```python

    # Cone half-angle from average slope
    avg_slope = (abs(m_upper) + abs(m_lower)) / 2.0
    cone_angle_rad = np.arctan(avg_slope)
    cone_angle_deg = np.degrees(cone_angle_rad)

    # Base height (at rightmost x position) - use actual edge points, not extrapolation
    x_max = min(upper["x"].max(), lower["x"].max())

    # Find actual edge points near the base (within 5 pixels of x_max)
    upper_base_points = upper[upper["x"] ≥ x_max - 5]
    lower_base_points = lower[lower["x"] ≥ x_max - 5]

    # Get actual y values at the base
    if len(upper_base_points) > 0 and len(lower_base_points) > 0:
        y_upper_base = upper_base_points["y"].mean()
        y_lower_base = lower_base_points["y"].mean()
        base_height_pixels = abs(y_upper_base - y_lower_base)
    else:
        # Fallback to extrapolation if no points found
        y_upper_base = m_upper * x_max + b_upper
        y_lower_base = m_lower * x_max + b_lower
        base_height_pixels = abs(y_upper_base - y_lower_base)

    return {
        "vertex_x": vertex_x,
        "vertex_y": vertex_y,
        "upper_slope": m_upper,
        "lower_slope": m_lower,
        "upper_intercept": b_upper,
        "lower_intercept": b_lower,
        "cone_angle_rad": cone_angle_rad,
        "cone_angle_deg": cone_angle_deg,
        "base_height_pixels": base_height_pixels,
    }


def rasmussen_pressure_coefficient(M, theta_c, gamma=GAMMA):
    """
    Calculate pressure coefficient using Rasmussen's approximation (Eq. 11).

    Cp/θc² ≈ 1 + [(γ+1)M²]/(θc²+2) + [(γ-1)M²]/(θc²+2)*ln[(γ+1)/2 + 1/(M²θc²)]

    Parameters:
    -----------
    M : float
        Freestream Mach number
    theta_c : float
```

```
        Cone half-angle [radians]
    gamma : float
        Specific heat ratio

    Returns:
    --------
    float : Cp (pressure coefficient)
    """
    theta_c_sq = theta_c**2

    term1 = 1.0
    term2 = ((gamma + 1.0) * M**2) / (theta_c_sq + 2.0)

    log_arg = (gamma + 1.0) / 2.0 + 1.0 / (M**2 * theta_c_sq)
    term3 = ((gamma - 1.0) * M**2) / (theta_c_sq + 2.0) * np.log(log_arg)

    Cp_over_theta_sq = term1 + term2 + term3
    Cp = Cp_over_theta_sq * theta_c_sq

    return Cp


def rasmussen_shock_angle(M, theta_c, gamma=GAMMA):
    """
    Calculate shock angle using Rasmussen's approximation (Eq. 10).

    sin(β)/sin(θc) ≈ √[(γ+1)/2 + 1/(M² sin²θc)]

    Parameters:
    -----------
    M : float
        Freestream Mach number
    theta_c : float
        Cone half-angle [radians]
    gamma : float
        Specific heat ratio

    Returns:
    --------
    float : β (shock angle) [radians]
    """
    sin_theta_c = np.sin(theta_c)

    term = (gamma + 1.0) / 2.0 + 1.0 / (M**2 * sin_theta_c**2)
    sin_beta = sin_theta_c * np.sqrt(term)

    beta = np.arcsin(sin_beta)

```

```python
    return beta


# --- Analysis ----------------------------------------------------------------

print("1. Calculating Freestream Conditions")
print("-" * 80)

# Step 1: Tube Mach number
Mt = calculate_mt_from_area_ratio(AREA_RATIO_TUBE, GAMMA)
print(f"  Mt (charge tube Mach number): {Mt:.6f}")

# Step 2: Expansion wave conditions
pt, Tt = calculate_expansion_conditions(Mt, P_FILL, T_FILL, GAMMA)
print(f"  pt (pressure behind expansion): {pt:.2f} Pa")
print(f"  Tt (temperature behind expansion): {Tt:.2f} K")

# Step 3: Stagnation conditions
p0, T0 = calculate_stagnation_conditions(Mt, pt, Tt, GAMMA)
print(f"  p0 (stagnation pressure): {p0:.2f} Pa")
print(f"  T0 (stagnation temperature): {T0:.2f} K")

# Step 4: Freestream conditions
freestream = calculate_freestream_conditions(M_EXPERIMENTAL, p0, T0, R_GAS, GAMMA)
print(f"  M∞ (freestream Mach number): {M_EXPERIMENTAL}")
print(f"  p∞ (freestream pressure): {freestream['p']:.2f} Pa")
print(f"  T∞ (freestream temperature): {freestream['T']:.2f} K")
print(f"  ρ∞ (freestream density): {freestream['rho']:.6f} kg/m³")
print(f"  a∞ (freestream sound speed): {freestream['a']:.2f} m/s")
print(f"  V∞ (freestream velocity): {freestream['V']:.2f} m/s")
print()

print("2. Loading and Processing Edge Detection Data")
print("-" * 80)

# Determine available frames
edges_dir = DATA_DIR / "edges"
frame_files = sorted(edges_dir.glob("frame*_upper.txt"))
frame_numbers = [int(f.stem.split("_")[0].replace("frame", "")) for f in frame_files]
n_frames = len(frame_numbers)

print(f"  Found {n_frames} frames with edge data")
print(f"  Frame range: {min(frame_numbers)} to {max(frame_numbers)}")
print()

print("3. Extracting Cone Geometry")
print("-" * 80)

```

```python
# Extract geometry from all frames
geometries = []
for frame_num in frame_numbers:
    upper, lower = load_edge_data(frame_num)
    geom = fit_cone_edges_and_find_vertex(upper, lower)
    geom["frame"] = frame_num
    geom["time"] = frame_num * DT
    geometries.append(geom)

geom_df = pd.DataFrame(geometries)

# Calculate median cone angle from all frames
median_cone_angle_deg = geom_df["cone_angle_deg"].median()
median_cone_angle_rad = geom_df["cone_angle_rad"].median()

# Use base height from FIRST 20 frames (cone is closest to camera, least perspective error)
initial_base_height_px = geom_df.iloc[:20]["base_height_pixels"].median()
base_radius_m = (initial_base_height_px / 2.0) * M_PER_PIXEL

print(
    f"  Median cone half-angle: {median_cone_angle_deg:.3f}° ({median_cone_angle_rad:.6f} rad)"
)
print(f"  Initial base diameter (first 20 frames): {initial_base_height_px:.2f} pixels")
print(f"  Base radius: {base_radius_m * 1000:.3f} mm ({base_radius_m:.6f} m)")
print(f"  Base area: {np.pi * base_radius_m**2:.6e} m²")
print()

# Store for later use
THETA_C = median_cone_angle_rad
R_BASE = base_radius_m
A_BASE = np.pi * R_BASE**2

print("4. Analyzing Cone Motion and Calculating Drag")
print("-" * 80)

# Extract vertex positions over time (convert to meters)
time_array = geom_df["time"].values
x_position_pixels = geom_df["vertex_x"].values
x_position_m = x_position_pixels * M_PER_PIXEL

# Fit quadratic polynomial: x = a0 + a1*t + a2*t^2
coeffs = np.polyfit(time_array, x_position_m, 2)
a2, a1, a0 = coeffs  # Note: polyfit returns highest degree first

print("  Quadratic fit coefficients:")
print(f"    a0 = {a0:.6f} m")
print(f"    a1 = {a1:.6f} m/s")
```

```python
print(f"    a2 = {a2:.6f} m/s²")
print()

# Calculate acceleration
ax_experimental = 2.0 * a2

print(f"  Experimental acceleration: ax = {ax_experimental:.3f} m/s²")
print()

# Calculate drag force
drag_force_experimental = CONE_MASS * ax_experimental

print(f"  Experimental drag force: D = {drag_force_experimental:.6f} N")
print()

# Calculate experimental drag coefficient
q_inf = 0.5 * freestream["rho"] * freestream["V"] ** 2  # Dynamic pressure
CD_experimental = drag_force_experimental / (q_inf * A_BASE)

print(f"  Dynamic pressure: q∞ = {q_inf:.2f} Pa")
print(f"  Experimental drag coefficient: CD = {CD_experimental:.6f}")
print()

# Generate fit curve for visualization
x_fit = a0 + a1 * time_array + a2 * time_array**2

print("5. Calculating Theoretical Drag (Rasmussen Approximation)")
print("-" * 80)

# Calculate Cp
Cp_theory = rasmussen_pressure_coefficient(M_EXPERIMENTAL, THETA_C, GAMMA)
print(f"  Rasmussen Cp (inviscid): {Cp_theory:.6f}")
print()

# For slender cones, Cp ≈ CD_inviscid
CD_inviscid = Cp_theory

# Calculate theoretical drag force
drag_force_inviscid = CD_inviscid * q_inf * A_BASE

print(f"  Inviscid drag coefficient: CD_inviscid = {CD_inviscid:.6f}")
print(f"  Inviscid drag force: D_inviscid = {drag_force_inviscid:.6f} N")
print()

# Calculate theoretical acceleration
ax_inviscid = drag_force_inviscid / CONE_MASS

print(f"  Inviscid acceleration: ax_inviscid = {ax_inviscid:.3f} m/s²")
```

```python
print()

print("6. Estimating Viscous Drag Component")
print("-" * 80)

# Viscous drag is the difference between experimental and inviscid
drag_force_viscous = drag_force_experimental - drag_force_inviscid
CD_viscous = CD_experimental - CD_inviscid

# Percentage breakdown
if drag_force_experimental ≠ 0:
    viscous_fraction = drag_force_viscous / drag_force_experimental * 100.0
    inviscid_fraction = drag_force_inviscid / drag_force_experimental * 100.0
else:
    viscous_fraction = 0.0
    inviscid_fraction = 0.0

print(f"  Total drag (experimental): {drag_force_experimental:.6f} N")
print(
    f"  Inviscid drag (theoretical): {drag_force_inviscid:.6f} N ({inviscid_fraction:.1f}%)"
)
print(
    f"  Viscous drag (estimated): {drag_force_viscous:.6f} N ({viscous_fraction:.1f}%)"
)
print()
print(f"  CD_experimental: {CD_experimental:.6f}")
print(f"  CD_inviscid: {CD_inviscid:.6f}")
print(f"  CD_viscous: {CD_viscous:.6f}")
print()

print("7. Calculating Theoretical Shock Angle")
print("-" * 80)

# Calculate shock angle
beta_rad = rasmussen_shock_angle(M_EXPERIMENTAL, THETA_C, GAMMA)
beta_deg = np.degrees(beta_rad)

print(f"  Cone half-angle: θc = {np.degrees(THETA_C):.3f}°")
print(f"  Shock angle: β = {beta_deg:.3f}° ({beta_rad:.6f} rad)")
print()

# --- Visualization ------------------------------------------------------------

print("8. Generating Figures")
print("-" * 80)

# Figure 1: Position vs Time
fig, ax = plt.subplots(figsize=(10, 6))
```

```python
ax.plot(
    time_array * 1000,
    x_position_m * 1000,
    "o",
    label="Measured vertex position",
    markersize=4,
    alpha=0.6,
)
ax.plot(
    time_array * 1000,
    x_fit * 1000,
    "-",
    label=f"Quadratic fit: $x = {a0:.3f} + {a1:.3f}t + {a2:.3f}t^2$",
    linewidth=2,
    color="red",
)
ax.set_xlabel("Time [ms]", fontsize=12)
ax.set_ylabel("Position [mm]", fontsize=12)
ax.set_title("Cone Vertex Position vs. Time", fontsize=14)
ax.legend(fontsize=10)
ax.grid(True, alpha=0.3)
plt.tight_layout()
plt.savefig(OUTPUT_FIGS_DIR / "position_vs_time.png", dpi=300)
plt.close()
print("  Saved: position_vs_time.png")

# Figure 2: Velocity vs Time
velocity_measured = np.gradient(x_position_m, time_array)
velocity_fit = a1 + 2 * a2 * time_array

fig, ax = plt.subplots(figsize=(10, 6))
ax.plot(
    time_array * 1000,
    velocity_measured,
    "o",
    label="Numerical derivative",
    markersize=3,
    alpha=0.5,
)
ax.plot(
    time_array * 1000,
    velocity_fit,
    "-",
    label=f"From fit: $v = {a1:.3f} + 2({a2:.3f})t$",
    linewidth=2,
    color="red",
)
ax.set_xlabel("Time [ms]", fontsize=12)
```

```python
ax.set_ylabel("Velocity [m/s]", fontsize=12)
ax.set_title("Cone Velocity vs. Time", fontsize=14)
ax.legend(fontsize=10)
ax.grid(True, alpha=0.3)
plt.tight_layout()
plt.savefig(OUTPUT_FIGS_DIR / "velocity_vs_time.png", dpi=300)
plt.close()
print("  Saved: velocity_vs_time.png")

# Figure 3: Acceleration vs Time
acceleration_measured = np.gradient(velocity_measured, time_array)

fig, ax = plt.subplots(figsize=(10, 6))
ax.plot(
    time_array * 1000,
    acceleration_measured,
    "o",
    label="From numerical derivatives",
    markersize=3,
    alpha=0.5,
)
ax.axhline(
    y=ax_experimental,
    color="red",
    linestyle="--",
    linewidth=2,
    label=f"From quadratic fit: $a_x = {ax_experimental:.3f}$ m/s²",
)
ax.set_xlabel("Time [ms]", fontsize=12)
ax.set_ylabel("Acceleration [m/s²]", fontsize=12)
ax.set_title("Cone Acceleration vs. Time", fontsize=14)
ax.legend(fontsize=10)
ax.grid(True, alpha=0.3)
plt.tight_layout()
plt.savefig(OUTPUT_FIGS_DIR / "acceleration_vs_time.png", dpi=300)
plt.close()
print("  Saved: acceleration_vs_time.png")

# Figure 4: Theoretical shock shape overlay on experimental image
# Select a frame near the middle with low angle of attack
middle_frame = frame_numbers[len(frame_numbers) // 2]
upper_mid, lower_mid = load_edge_data(middle_frame)
geom_mid = geom_df[geom_df["frame"] == middle_frame].iloc[0]

# Load the actual shadowgraph image
image_path = DATA_DIR / "images" / f"cone_{middle_frame:03d}.tif"
img = Image.open(image_path)
img_array = np.array(img)
```

```python

fig, ax = plt.subplots(figsize=(14, 10))

# Display the shadowgraph image
ax.imshow(img_array, cmap="gray", alpha=0.8, extent=[0, img_array.shape[1], img_array.shape[0], 0])

# Plot cone edges
ax.plot(upper_mid["x"], upper_mid["y"], "b-", linewidth=2.5, label="Detected cone edges")
ax.plot(lower_mid["x"], lower_mid["y"], "b-", linewidth=2.5)

# Plot vertex
vx, vy = geom_mid["vertex_x"], geom_mid["vertex_y"]
ax.plot(vx, vy, "go", markersize=12, label="Cone vertex", zorder=5, markeredgecolor="white", markeredgewidth=1.5)

# Plot theoretical shock rays
x_shock_range = np.linspace(vx, upper_mid["x"].max(), 200)

# Upper shock: y = vy + (x - vx) * tan(β)
y_shock_upper = vy + (x_shock_range - vx) * np.tan(beta_rad)

# Lower shock: y = vy - (x - vx) * tan(β)
y_shock_lower = vy - (x_shock_range - vx) * np.tan(beta_rad)

ax.plot(
    x_shock_range,
    y_shock_upper,
    "r--",
    linewidth=3,
    label=f"Theoretical shock (β = {beta_deg:.2f}°)",
)
ax.plot(x_shock_range, y_shock_lower, "r--", linewidth=3)

# Annotations
ax.text(
    vx + 50,
    vy + 60,
    f"θc = {np.degrees(THETA_C):.2f}°",
    fontsize=12,
    bbox=dict(boxstyle="round", facecolor="yellow", alpha=0.8),
    fontweight="bold",
)
ax.text(
    vx + 100,
    vy + (x_shock_range[50] - vx) * np.tan(beta_rad) + 15,
    f"β = {beta_deg:.2f}°",
    fontsize=12,
```

```python
    color="red",
    bbox=dict(boxstyle="round", facecolor="white", alpha=0.9),
    fontweight="bold",
)

ax.set_xlabel("x [pixels]", fontsize=13)
ax.set_ylabel("y [pixels]", fontsize=13)
ax.set_title(f"Theoretical Shock Overlay on Experimental Shadowgraph (Frame
↪ {middle_frame})", fontsize=14, fontweight="bold")
ax.legend(fontsize=11, loc="upper left", framealpha=0.9)
ax.set_xlim(0, img_array.shape[1])
ax.set_ylim(img_array.shape[0], 0)
plt.tight_layout()
plt.savefig(OUTPUT_FIGS_DIR / "theoretical-shock.png", dpi=300, bbox_inches="tight")
plt.close()
print("  Saved: theoretical-shock.png (updated with shadowgraph background)")

# Figure 5: Drag coefficient comparison bar chart
fig, ax = plt.subplots(figsize=(8, 6))

categories = ["Experimental\n(Total)", "Inviscid\n(Theory)", "Viscous\n(Estimated)"]
values = [CD_experimental, CD_inviscid, CD_viscous]
colors = ["#1f77b4", "#ff7f0e", "#2ca02c"]

bars = ax.bar(categories, values, color=colors, alpha=0.7, edgecolor="black")

# Add value labels on bars
for bar, val in zip(bars, values):
    height = bar.get_height()
    ax.text(
        bar.get_x() + bar.get_width() / 2.0,
        height,
        f"{val:.4f}",
        ha="center",
        va="bottom",
        fontsize=11,
        fontweight="bold",
    )

ax.set_ylabel("Drag Coefficient", fontsize=12)
ax.set_title("Drag Coefficient Comparison", fontsize=14)
ax.grid(True, axis="y", alpha=0.3)
plt.tight_layout()
plt.savefig(OUTPUT_FIGS_DIR / "drag_coefficient_comparison.png", dpi=300)
plt.close()
print("  Saved: drag_coefficient_comparison.png")

# Figure 6: Cone angle variation (quality check)
```

```python
fig, ax = plt.subplots(figsize=(10, 6))
ax.plot(geom_df["time"] * 1000, geom_df["cone_angle_deg"], "o-", markersize=3)
ax.axhline(
    y=median_cone_angle_deg,
    color="red",
    linestyle="--",
    linewidth=2,
    label=f"Median: {median_cone_angle_deg:.3f}°",
)
ax.set_xlabel("Time [ms]", fontsize=12)
ax.set_ylabel("Measured Cone Half-Angle [°]", fontsize=12)
ax.set_title("Cone Half-Angle Over Time (Measurement Consistency)", fontsize=14)
ax.legend(fontsize=10)
ax.grid(True, alpha=0.3)
plt.tight_layout()
plt.savefig(OUTPUT_FIGS_DIR / "cone_angle_consistency.png", dpi=300)
plt.close()
print("  Saved: cone_angle_consistency.png")

print()

# --- Output ------------------------------------------------------------------

print("9. Exporting Data to CSV")
print("-" * 80)

# Table 1: Freestream conditions
freestream_data = {
    "Parameter": [
        "M_t",
        "p_t [Pa]",
        "T_t [K]",
        "p_0 [Pa]",
        "T_0 [K]",
        "M_∞ [experimental]",
        "p_∞ [Pa]",
        "T_∞ [K]",
        "ρ_∞ [kg/m³]",
        "a_∞ [m/s]",
        "V_∞ [m/s]",
    ],
    "Value": [
        Mt,
        pt,
        Tt,
        p0,
        T0,
        M_EXPERIMENTAL,
```

```
        freestream["p"],
        freestream["T"],
        freestream["rho"],
        freestream["a"],
        freestream["V"],
    ],
}
df_freestream = pd.DataFrame(freestream_data)
df_freestream.to_csv(OUTPUT_TEXT_DIR / "freestream_conditions.csv", index=False)
print("  Saved: freestream_conditions.csv")

# Table 2: Cone geometry
cone_geom_data = {
    "Parameter": [
        "Half-angle [rad]",
        "Half-angle [°]",
        "Base radius [m]",
        "Base radius [mm]",
        "Base area [m²]",
    ],
    "Value": [
        THETA_C,
        np.degrees(THETA_C),
        R_BASE,
        R_BASE * 1000,
        A_BASE,
    ],
}
df_cone_geom = pd.DataFrame(cone_geom_data)
df_cone_geom.to_csv(OUTPUT_TEXT_DIR / "cone_geometry.csv", index=False)
print("  Saved: cone_geometry.csv")

# Table 3: Drag analysis summary
drag_summary_data = {
    "Parameter": [
        "Quadratic fit a0 [m]",
        "Quadratic fit a1 [m/s]",
        "Quadratic fit a2 [m/s²]",
        "Experimental acceleration [m/s²]",
        "Experimental drag force [N]",
        "Experimental drag coefficient",
        "Rasmussen Cp (inviscid)",
        "Inviscid drag force [N]",
        "Inviscid drag coefficient",
        "Viscous drag force [N]",
        "Viscous drag coefficient",
        "Viscous fraction [%]",
    ],
```

```
    "Value": [
        a0,
        a1,
        a2,
        ax_experimental,
        drag_force_experimental,
        CD_experimental,
        Cp_theory,
        drag_force_inviscid,
        CD_inviscid,
        drag_force_viscous,
        CD_viscous,
        viscous_fraction,
    ],
}
df_drag = pd.DataFrame(drag_summary_data)
df_drag.to_csv(OUTPUT_TEXT_DIR / "drag_analysis.csv", index=False)
print("  Saved: drag_analysis.csv")

# Table 4: Shock angle
shock_data = {
    "Parameter": [
        "Cone half-angle [rad]",
        "Cone half-angle [°]",
        "Shock angle [rad]",
        "Shock angle [°]",
    ],
    "Value": [THETA_C, np.degrees(THETA_C), beta_rad, beta_deg],
}
df_shock = pd.DataFrame(shock_data)
df_shock.to_csv(OUTPUT_TEXT_DIR / "shock_angle.csv", index=False)
print("  Saved: shock_angle.csv")

# Table 5: Full trajectory data
geom_df.to_csv(OUTPUT_TEXT_DIR / "trajectory_data.csv", index=False)
print("  Saved: trajectory_data.csv")

print()
print("=" * 80)
print("Analysis Complete!")
print("=" * 80)
```